



Ruby

Building Elegant and Efficient Applications with Ruby

Ruby is a dynamic, open-source programming language with a focus on simplicity and productivity. Known for its elegant syntax and readability, Ruby has become a favorite among developers for web development, particularly with the Ruby on Rails framework.

Introduction to Ruby

Ruby was created by Yukihiro “Matz” Matsumoto in the mid-1990s. It combines parts of his favorite languages (Perl, Smalltalk, Eiffel, Ada, and Lisp) to form a new language that balances functional and imperative programming. Ruby’s design emphasizes developer happiness and productivity, with an elegant syntax that is natural to read and easy to write.

Key Features of Ruby:

- **Object-Oriented:** Everything in Ruby is an object, including

primitive data types.

- **Dynamic Typing and Duck Typing:** Ruby uses dynamic typing and duck typing, making it flexible and allowing for rapid development.
- **Elegant Syntax:** Ruby’s syntax is clean and readable, resembling natural language.
- **Garbage Collection:** Automatic memory management through garbage collection.
- **Rich Standard Library:** Includes a vast collection of built-in functions and libraries.
- **Metaprogramming:** Allows writing code that writes code, making it highly extensible.

Setting Up Ruby

Before you start coding in Ruby, you need to set up your development environment. Here’s how you can get started: